Projective resolutions, surface orders and zeta determinants

A constructive framework and the arithmetic comparison problem

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Abstract

We develop the homological interface in a proposed surface-order approach to the generalized Riemann hypothesis. Forbidden-arrow chains give explicit minimal resolutions, Ext and Tor; permitted-arrow states give a short proof of the weighted Cartan determinant formula. We distinguish these formal invariants from determinant lines and Frobenius zeta functions, correct periodicity, grading and base-change errors, and state a precise conditional arithmetic program. The algebraic results have complete proofs; selected matrix and module components are checked in Lean4. The missing number-field determinant and positive realization remain open. This is not a proof of GRH.

Contents

1 Objects, conventions and scope **1**

2 Minimal projective resolutions and the signed Euler matrix **2**

3 The weighted Cartan determinant from arrow states **3**

4 Completion, singular seams and geometric objects **4**

5 Determinant lines, chain endomorphisms and zeta functions **5**

6 What finite-field purity supplies **6**

7 A precise arithmetic program and a closure theorem **7**

8 Verification and reproducibility **8**

1 Objects, conventions and scope

The starting sources are Schreiber's surface-algebra and surface-order papers [1, 2], the generic-resolution discussion [3], and the weighted locally gentle determinant formula of Bessenrodt–Holm [4]. Our purpose is to supply the exact homological constructions needed to use their useful parts in a Hilbert–Pólya program. The source manuscripts are preserved; corrections below are statements in this new account. Full reading records, pinned versions and artifact hashes accompany the project. Related literature is used with the hypotheses stated here, rather than incorporated as an unqualified equivalence.

Fix a field k, a finite quiver $Q = (Q_0, Q_1, s, t)$, and an ideal I generated by a set of length-two paths. We compose paths in the order traveled: ab means first a, then b. Write $A = kQ/I$, e_i for its vertex idempotents, and $J = A_{>0}$ for the arrow ideal. A path is *permitted* when none of its adjacent pairs belongs to I. These paths, including the empty paths e_i, form a k-basis of A. The algebra may be infinite dimensional.

Definition 1.1. The pair (Q, I) is locally gentle if each vertex has at most two incoming and two outgoing arrows, and every arrow has at most one permitted successor, one forbidden successor, one permitted predecessor and one forbidden predecessor. A gentle algebra is additionally finite dimensional. We use right modules, $P_i = e_iA$ and $S_i = P_i/P_iJ$, with arrows of degree one and internal shift $(M\langle n\rangle)_d = M_{d-n}$.

Thus $P_i\langle n\rangle$ is generated in degree n; square brackets later denote the cohomological shift in a derived category. Completion $\widehat{A} = \varprojlim A/J^r$, a hereditary normalization of a completed order, and a central special fiber are different objects. A forbidden chain is a composable sequence all of whose adjacent pairs lie in I; it is generally zero as a path in A. We retain it as an indexing word. Forgetting this distinction makes the resolution below disappear.

The number-field target is initially primitive Dirichlet $L(s, \chi)$. Finite-order Hecke characters over a specified number field, Dedekind zeta functions, and general Artin representations require separate comparison theorems. General Artin holomorphy is not assumed. None of the algebraic results below identifies its determinant with any of these functions.

2 Minimal projective resolutions and the signed Euler matrix

For $n \geq 1$, let $\mathcal{R}_n(i)$ be the forbidden chains of length n starting at i; set $\mathcal{R}_0(i) = \{e_i\}$. For a nonempty chain $p = a_1 \cdots a_n$, write $p^- = a_1 \cdots a_{n-1}$.

Theorem 2.1 (Explicit linear resolution)**.** *For every quadratic monomial algebra as above, a graded projective resolution of S_i is*

$$\cdots \longrightarrow P_n(i) \xrightarrow{d_n} P_{n-1}(i) \longrightarrow \cdots \longrightarrow P_0(i) \longrightarrow S_i \longrightarrow 0, \qquad P_n(i) = \bigoplus_{p \in \mathcal{R}_n(i)} e_{t(p)}A\langle n\rangle.$$

On the p summand, $d_n(x) = a_n x$ in the p^- summand. The augmentation is the quotient $e_iA \to S_i$. The maps are right A-linear, homogeneous of degree zero, and valued in the graded arrow radical. The resolution is minimal in the graded sense. In particular A with its natural path grading is Koszul.

Proof. Each summand is an idempotent projective. The endpoints make left multiplication $a_n x$ a map into the indicated right module; associativity proves right linearity. The source generator has degree n, and its image has degree $1 + (n-1) = n$. Two successive maps multiply by $a_{n-1}a_n = 0$.

The augmentation kernel is the span of all nonempty permitted paths from i, the direct sum of aA over arrows a from i. This is $\operatorname{im} d_1$. More generally, the kernel of $e_{t(a)}A \xrightarrow{x \mapsto ax} e_{s(a)}A$ is the span of permitted paths whose first arrow b satisfies $ab \in I$. Indeed $ae_{t(a)} = a \neq 0$; every other basis path either vanishes after multiplication because of that first pair, or maps to a distinct surviving basis path. The kernel is therefore $\bigoplus_{b:ab\in I} bA$. For a fixed prefix p^-, images of distinct last arrows have disjoint nonzero path supports. There can be no cancellation between them. Taking their individual kernels identifies $\ker d_n$ with the image of the summands indexed by pb. This proves exactness in every degree without a global-dimension assumption. All entries have positive degree (graded

minimality does not identify J with the ordinary Jacobson radical), so applying $\mathrm{Hom}_{\text{gr-}A}(-, S_j\langle r\rangle)$ kills the differentials. Minimality and linearity follow. The direct sum of the S_i is A/J, so its linear resolution is the Koszul property. □

Corollary 2.2 (Ext, Tor and inversion). In ungraded modules, $\dim_k \mathrm{Ext}^n_A(S_i, S_j)$ is the number of forbidden chains of length n from i to j. The same number is $\dim_k \mathrm{Tor}^A_n(S_i, {}_jS)$, where ${}_jS = Ae_j/JAe_j$. With the above grading, Ext is supported on the matching internal shift n. Define

$$C_{ij}(z) = \sum_{d\geq 0} \dim_k(e_i A_d e_j) z^d, \qquad E_{ij}(z) = \sum_{n\geq 0} (-1)^n \dim_k \mathrm{Ext}^n_A(S_i, S_j) z^n.$$

Then $E(z)C(z) = C(z)E(z) = 1$ in $M_{Q_0}(\mathbb{Z}[[z]])$.

Proof. An A-map $e_vA \to S_j$ is determined by an element of S_je_v, of dimension δ_{vj}. Each differential becomes zero since arrows annihilate the simple. Tensoring with a left simple gives the same calculation. Degree by degree, the augmented resolution has Euler characteristic S_i. Its nth term contributes $(-1)^n z^n C_{t(p),j}(z)$, so the alternating sum is exactly $(EC)_{ij} = \delta_{ij}$. Each coefficient is finite: $n \leq d$ and Q is finite. Since both matrices have constant term the identity, the formal matrix inverse is unique and two-sided. □

This already signed E is evaluated at z, not at $-z$. Other conventions can transpose the matrices, but cannot justify applying the sign twice. The complementary-relation algebra $A^{\#} = kQ/I^{\#}$ allows precisely the forbidden pairs of A, and $E(z) = C_{A^{\#}}(-z)$. Its paths encode the quadratic Koszul dual, with an opposite quiver if that convention is used for Yoneda composition. At the chain level Yoneda products are concatenation of forbidden chains (zero when the join is permitted), with the handedness/opposite convention just declared. An explicit comparison-map proof is given in the extended edition; the dimensions and inversion above do not require this product identification. Thus endpoint data and relations, rather than Betti numbers alone, control the product.

For locally gentle A, each initial arrow has at most one forbidden continuation. The resolution has at most two arms; they terminate or enter a finite cycle. A repeated ordered arrow pattern yields an eventual tail period after forgetting internal grading. This does not say that $\Omega^p S_i \cong S_i$, or that the displayed period is the least module-isomorphism period.

3 The weighted Cartan determinant from arrow states

Give each arrow an independent commuting weight w_a and work over $R = \mathbb{Z}[[w_a : a \in Q_1]]$. A path has weight the product of its arrows. The weighted Cartan matrix $C_{ij}(\mathbf{w})$ sums permitted paths from i to j. Finite Q guarantees finitely many terms of each total degree.

Put $n = |Q_0|$, $m = |Q_1|$ and define

$$S_{i,a} = \mathbf{1}_{s(a)=i}, \quad T_{a,j} = \mathbf{1}_{t(a)=j}, \quad D_{a,b} = \delta_{ab} w_a,$$

$$U_{a,b} = \mathbf{1}_{t(a)=s(b),\, ab\notin I} w_b, \qquad V_{a,b} = \mathbf{1}_{t(a)=s(b),\, ab\in I} w_b.$$

Here S is $n \times m$, T is $m \times n$, and U, V, D are $m \times m$. Every composable pair is exactly one kind, so $U + V = TSD$.

Theorem 3.1 (Weighted cycle formula, cf. Bessenrodt–Holm)**.** *For a finite locally gentle quiver,*

$$\det C_A(\mathbf{w}) = \frac{\displaystyle\prod_{c\in\mathcal{C}_{\mathrm{forb}}} (1 - (-1)^{\ell(c)} w(c))}{\displaystyle\prod_{c\in\mathcal{C}_{\mathrm{perm}}} (1 - w(c))}.$$

Both sets consist of primitive directed cycles of the corresponding arrow-transition graph, modulo cyclic rotation. Iterations are not extra factors. All denominators have constant term one.

Proof. Paths of length $r+1$ contribute SDU^rT. Hence

$$C = 1 + SD(1-U)^{-1}T, \qquad (1-U)^{-1} = \sum_{r\geq 0} U^r.$$

The inverse converges in the augmentation topology because every entry of U has positive degree. Sylvester's determinant identity $\det(1+AB) = \det(1+BA)$ for rectangular A, B gives

$$\det C = \det\bigl(1 + (1-U)^{-1}TSD\bigr) = \frac{\det(1+V)}{\det(1-U)}.$$

For completeness the rectangular identity follows by expanding the determinant of the block matrix $\left(\begin{smallmatrix} 1 & -A \\ B & 1 \end{smallmatrix}\right)$ with either identity block. Thus it is valid over any commutative ring. Multiplying before division gives a cleared identity even in more general coefficient rings.

Each vertex of the U transition graph has at most one predecessor and one successor, so every connected component is a directed chain or a directed cycle. The same holds for V. Ordering a chain along its arrows makes its transition matrix strictly triangular, contributing determinant one. On a cycle of length r, only the identity permutation and the full cyclic permutation contribute to $\det(1-U)$; their sum is $1 - w(c)$. Replacing U by $-V$ gives $1 - (-1)^r w(c)$. For a one-loop component this is simply $1 - w_a$ or $1 + w_a$, so there is no exceptional sign case. Multiplying the component determinants proves the formula, including quivers with both infinite permitted paths and infinite resolutions. □

The transfer identity before component factorization holds for every finite quadratic monomial quiver. The locally gentle hypothesis is used for the chain/cycle decomposition, not for Sylvester's identity. Formula 3.1 is a formal invariant, not the determinant of an infinite resolution. Specializing weights to a point where a denominator vanishes requires a separate cancellation or continuation argument.

4 Completion, singular seams and geometric objects

The node illustrates both the strength and the limits of the construction. For $A = k\langle x, y\rangle/(xy, yx) = k[x, y]/(xy)$,

$$C_A(w_x, w_y) = \frac{1 - w_x w_y}{(1-w_x)(1-w_y)}, \quad C_A(z) = \frac{1+z}{1-z}, \quad E_A(z) = \frac{1-z}{1+z}.$$

The completed order $\Lambda = k[[x, y]]/(xy)$ has normalization $B = k[[x]] \times k[[y]]$ and conductor square

$$\Lambda = B \times_{k\times k} k,$$

where the lower map is diagonal and $B \to k \times k$ evaluates the two origins. Equality of constant terms proves this description: (f, g) corresponds uniquely to $f(x) + g(y) - f(0)$. No residue map $k((x)) \to k$ is involved; x is a unit in that field.

Its simple k has resolution

$$\cdots \xrightarrow{\mathrm{diag}(x,y)} \Lambda^2 \xrightarrow{\mathrm{diag}(y,x)} \Lambda^2 \xrightarrow{[x\ y]} \Lambda \longrightarrow k \longrightarrow 0.$$

Indeed $\operatorname{ann}(x) = (y)$, $\operatorname{ann}(y) = (x)$ and $(x) \cap (y) = 0$. With $t = x + y$, Λ is free of rank two over $k[[t]]$, whereas k is t-torsion. Positive syzygies are torsion-free over this DVR. Thus the simple is not periodic from degree zero. Moreover $(x) \oplus (y)$ and $(y) \oplus (x)$ are isomorphic: the unlabelled syzygy tail has period one even though the ordered branch pattern has period two. Tensoring with $k[[t]]/(t)$ does not preserve this resolution: the special fiber is $k[x]/(x^2)$, whose simple has Betti numbers $1, 1, 1, \ldots$, rather than $1, 2, 2, \ldots$. The missing homology is a Tor contribution.

For the following closed split subclass, require exactly two incoming and two outgoing arrows at each vertex, unique permitted and forbidden successors and predecessors, and two matched permitted-cycle incidences at each original vertex, with no unmatched endpoints. One can then specify hereditary cycle orders H_c over $k[[x_c]]$ and impose equality of the paired diagonal residues. Map each permitted cycle to successive matrix units, with its wrap arrow multiplied by x_c. Distinct permitted paths have independent coefficients in these entries, and every tuple with matching diagonal residues is reconstructed coefficient by coefficient. This proves the completed path-algebra/pullback comparison in this split setting. A common central parameter is $t \mapsto (x_c 1)_c$; long permitted paths show that the arrow-adic and t-adic filtrations are cofinal. Broader Dedekind bases need their own residue identifications, local uniformizers and projective-cover hypotheses. Hereditary normalization does not make the glued order hereditary.

The derived category $D(A)$ is obtained from complexes of right A-modules by inverting quasi-isomorphisms. Its perfect subcategory $\operatorname{Perf}(A)$ consists of objects represented by bounded complexes of finitely generated projectives. A simple with an infinite minimal resolution need not be perfect. The homotopy category of bounded projectives embeds as $\operatorname{Perf}(A)$; $D^b(\text{mod-}A)$ can be larger. This is the correct finiteness test before forming finite determinants.

For finite-dimensional graded gentle algebras, the surface models of [6, 7, 8, 9] describe string objects by graded curves and band objects by closed curves with local systems. A line field/degree convention and endpoint or puncture behavior are essential. Appropriate oriented graded intersections give morphism bases, subject to exceptional closed-curve identity and Auslander–Reiten maps. Thus a bare count of transverse intersections is not a universal Hom formula. In a module heart, $\operatorname{Ext}^n(M, N) = \operatorname{Hom}_{D^b}(M, N[n])$; resolutions compute these groups and derived tensor products compute Tor. Rotating or zigzagging curves supplies concrete projective covers in the stated models. Koszul duality exchanges permitted and forbidden combinatorics through a derived functor with finiteness/completion conditions; a formal matrix inverse alone is not a derived equivalence. There is no justified transfer from these finite-dimensional categories to every completed arithmetic order, nor an identification of curve length with prime norm.

5 Determinant lines, chain endomorphisms and zeta functions

For a bounded complex $P^\bullet$ of finite projective modules over a commutative base R, its determinant is the graded invertible module

$$\det(P^\bullet) = \bigotimes_i (\det_R P^i)^{(-1)^i}.$$

It is a line, with the coherent signs of the determinant functor, not a scalar analytic function. Quasi-isomorphism invariance and derived base change hold in the perfect category [17, 18, 19]. For noncommutative A first choose a realization to a perfect complex over a commutative base; arbitrary A-projectives have no ordinary top-exterior-power determinant of this kind. A resolution is a computational presentation, not an additional arithmetic eigenvalue source.

Theorem 5.1 (Cohomological cancellation)**.** *Let $C^\bullet$ be a bounded finite-dimensional complex over a field and let F be a chain endomorphism. Then*

$$Z(C,F;u) := \prod_i \det(1-uF \mid C^i)^{(-1)^{i+1}} = \prod_i \det(1-uF \mid H^i(C))^{(-1)^{i+1}}.$$

An acyclic complex contributes one.

Proof. Set $B^i = \operatorname{im} d^{i-1}$ and $Z^i = \ker d^i$. Commutation with d preserves both. The two exact sequences $0 \to Z^i \to C^i \to B^{i+1} \to 0$ and $0 \to B^i \to Z^i \to H^i \to 0$ make F block triangular in adapted bases. Hence its characteristic determinant on C^i is the product of those on B^i, H^i, B^{i+1}. The two boundary factors have opposite adjacent exponents and cancel. The finite product requires no convergence argument. □

An infinite periodic resolution is not a bounded complex for this theorem. A truncation leaves a boundary syzygy; omitting it changes the object. Likewise the matrices U and $-V$ in Theorem 3.1 can package a formal virtual determinant, but have not been shown to be an arithmetic endomorphism commuting with the resolution differential. These are separate constructions.

Over a characteristic-zero field (or after extending scalars to a $\mathbb{Q}$-algebra), for a finite matrix M, differentiation of the determinant at the identity and the formal inverse series give

$$-\log\det(1-uM) = \sum_{r\geq 1} \frac{\operatorname{tr}(M^r)}{r} u^r.$$

Indeed the derivative of the left side is $\operatorname{tr}(M(1-uM)^{-1})$; expand and integrate term by term with constant term zero. For a partial permutation this becomes the primitive-cycle Euler product. Such transfer zeta functions are not automatically Hasse–Weil or number-field L-functions. An infinite-dimensional extension needs a stated Fredholm or regularized determinant and its convergence theorem.

6 What finite-field purity supplies

For a separated finite-type $X/\mathbb{F}_q$ and a suitable constructible ℓ-adic coefficient system $\mathcal{F}$, $\ell \neq \operatorname{char}\mathbb{F}_q$, the Grothendieck trace formula identifies

$$L(X,\mathcal{F},u) = \prod_{x \text{ closed}} \det(1-u^{\deg x}\operatorname{Fr}_x \mid \mathcal{F}_x)^{-1} = \prod_i \det(1-u\operatorname{Fr} \mid H^i_c(X_{\overline{\mathbb{F}}_q},\mathcal{F}))^{(-1)^{i+1}}.$$

We choose the Frobenius convention giving eigenvalue q on $H^2(\mathbb{P}^1)$. For a smooth projective geometrically connected curve and constant coefficients this is

$$Z(X,u) = \frac{\det(1-u\operatorname{Fr} \mid H^1)}{(1-u)(1-qu)}.$$

Deligne's purity theorem gives every complex absolute value $\sqrt{q}$ on H^1 [20, 21]; with $u = q^{-s}$ the numerator zeros have $\operatorname{Re} s = 1/2$. This paragraph invokes deep external arithmetic theorems, not consequences of the elementary path-resolution proof. Weil II also treats morphisms over $\mathbb{Z}$ with fiberwise weights; its scope is not confined to one finite field. What is absent here is a global completed number-field determinant and a positivity theorem for its zero-bearing part. Open or singular spaces require mixed weights or intersection complexes: for example $Z(\mathbb{G}_m,u) = (1-u)/(1-qu)$ has a weight-zero zero.

One useful finite-field test is $X : y^2 = x^3 - x$ over $\mathbb{F}_5$. Its eight points give trace -2 and

$$Z(X,u) = \frac{1 + 2u + 5u^2}{(1-u)(1-5u)}, \qquad \alpha_1, \alpha_2 = -1 \pm 2i,$$

with $|\alpha_i| = \sqrt{5}$. This benchmarks a genuine cohomological determinant. A nodal or gentle category is not automatically smooth and proper over k, the hypotheses of modern noncommutative Weil frameworks [22]. Surface topology alone supplies neither point counts nor this Frobenius realization.

Proposition 6.1 (Duality versus positive purity)**.** *An alternating duality $F^t J F = qJ$ does not force eigenvalues of F to have modulus $\sqrt{q}$. A positive Hermitian form H with $F^* H F = qH$, $q > 0$, does force that conclusion.*

Proof. For the first assertion take $q = 2$, $F = \operatorname{diag}(2, 1)$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Direct multiplication gives the duality, but its eigenvalues have squares of absolute values 4 and 1. For the second, if $Fv = \alpha v$ and $v \neq 0$, then $|\alpha|^2 v^* H v = (Fv)^* H (Fv) = q v^* H v$; divide by $v^* H v > 0$. □

Functional equations similarly pair zeros without placing them on a line. Local weight zero for finite-image Artin Frobenius, already known since its eigenvalues are roots of unity, is insufficient for global number-field GRH.

7 A precise arithmetic program and a closure theorem

Proposition 7.1 (Positive-length Cartan obstruction)**.** *In Theorem 3.1, specialize $w_a(s) = \eta_a e^{-s\lambda_a}$ with $|\eta_a| \leq 1$ and $\lambda_a > 0$. Its rational Cartan determinant and its reciprocal are holomorphic and nonzero on $\operatorname{Re} s > 0$. Neither, multiplied by holomorphic nonvanishing factors, can converge locally uniformly to a nonzero holomorphic target having a zero there.*

Proof. Every primitive cycle has positive total length and product weight of modulus strictly less than one in this half-plane. Each numerator and denominator factor is therefore nonzero; their finite product is holomorphic with holomorphic reciprocal. If a sequence of these functions converged locally uniformly to a nonzero target with a zero, choose a small closed disk around an isolated target zero and no boundary zero. Uniform boundary convergence and Rouché's theorem force an approximant zero in the disk, a contradiction. The identical argument handles the stated normalizations. □

Thus the resolution/Cartan formula rigorously diagnoses a failed direct Euler-product repair. Artificially centering the weights at $s - 1/2$ puts their elementary cycle zeros on a chosen line but proves no matching with an arithmetic L-function.

Theorem 7.2 (Conditional positive determinant closure)**.** *Let Ω be a connected open subset of $\mathbb{C}$ containing an open portion of the critical strip, and let $L_\circ$ be holomorphic on Ω, not identically zero. Suppose holomorphic functions D_N converge locally uniformly to $L_\circ$ and every zero of D_N in Ω lies on $\operatorname{Re} s = 1/2$. Then every zero of $L_\circ$ in Ω lies on that line. One sufficient finite realization is*

$$D_N(s) = g_N(s) \det(A_N - [-i(s - 1/2)]B_N), \quad A_N = A_N^*, \quad B_N > 0,$$

with g_N holomorphic and nonzero and the same convergence hypothesis.

Proof. At a singular pencil, a nonzero v satisfies $A_N v = z B_N v$. The quotient $z = (v^* A_N v)/(v^* B_N v)$ is real since its denominator is positive. For $z = -i(s - 1/2)$ this means $\operatorname{Re} s = 1/2$. If the limit had an off-line zero, a small disk avoiding the line and other zeros would contradict Rouché exactly as in Proposition 7.1. □

This theorem is a closure mechanism, not a construction of A_N, B_N, g_N. Our existing arithmetic partial-sum approximants converge with compact error bounds; no positive-pencil determinant identity for them has been proved. Replacing their matrices by Hermitian parts can change the determinant and is not such an identity.

A complete attempt using resolutions must discharge the following sequence of concrete claims.

1. Construct the arithmetic category and realization independently of unknown zeros. Specify coefficients, ramification, actual Galois/Frobenius action, archimedean data and conductor. A dessin's topological monodromy is not this action on one fixed algebra.

2. Choose a finite perfect realization, or an infinite analytic complex with a justified determinant. Prove chain-map compatibility and comparison through conductor gluing and derived base change; include boundary and singularity contributions. Retain determinant trivializations and normalization.

3. Prove the completed L-function identity, including gamma factors and multiplicities. Finite-dimensional finite-field rationality in u does not provide a finite-dimensional global Frobenius for primes of different norms. Number-field flow proposals [23] require a regularized determinant and a global generator.

4. Construct and prove positivity/polarization for the *same* zero-bearing realization, or prove a determinant-preserving positive-pencil comparison for the convergent arithmetic approximants. Alternating duality, local unitarity, positivity of unrelated moments and a formal Euler identity do not discharge this step.

5. Prove trace-norm or compact-uniform determinant convergence on an exhaustion, with a nonzero anchor and appropriate pole removal. Then apply Theorem 7.2, with each intended L-function scope and all exceptional points checked.

The local homological part can now be implemented explicitly. The global arithmetic identity, its positive realization and its analytic comparison remain mathematical research problems. No assumption of GRH, Artin holomorphy or unknown zero locations is used in the proved algebraic statements.

8 Verification and reproducibility

The separate pinned Lean4.24/mathlib layer verifies a general cleared version of the rectangular determinant identity, acyclic-pair characteristic-determinant cancellation, the actual alternating-duality counterexample, and exactness/projectivity components of a dual-number resolution over an actual nonsemisimple algebra. Existing files verify positive Hermitian pencils and analytic nonvanishing components. Theorems print their types and foundation dependencies. These are components, not a formalization of all forbidden-chain resolutions, the derived surface models, the trace formula, Deligne purity, or GRH.

A standard-library exact rational checker tests finite quivers, path/chain coefficients, differentials and bounded internal-degree exactness, signed matrix inversion and the weighted cycle determinant, including nodal, terminating, periodic and cancellation examples. Finite tests supplement the universal proofs; they do not certify the arithmetic bridge. All source submodules and pinned environments are preserved. Both editions compile from shared core text; the extended edition adds explicit computations, geometric hypotheses and implementation tasks.

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